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Hemp flowers cultivated on a soil contaminated with cadmium, lead and zinc exhibit valorization potential

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ABSTRACT

Industrial hemp (*Cannabis sativa L.*) is an effective crop for the phytomanagement of lands contaminated with trace metals, demonstrating tolerance to metal toxicity without significant impacts on its value-chain. However, the effects of metal contamination on hemp flowers remain understudied, limiting the assessment of the valorization potential for this valuable plant part. In a greenhouse experiment, we evaluated flower production, cannabinoid content and metal accumulation in hemp grown on a soil contaminated with Cd, Pb and Zn (pseudo-total concentrations: 13.0, 664 and 1048 mg kg^{-1}, respectively). Our results suggest a limited capacity for phytoextraction, with low removal rates for all three metals. Still, hemp flowers presented favorable features that support valorization potential. Both flower biomass and the synthesis of cannabidiol and tetrahydrocannabinol were comparable to plants grown under reference conditions. Notably, inflorescences exhibited the lowest accumulation of Cd and Pb among all plant tissues. Concentrations for these elements were 0.45 and 1.1 mg kg^{-1} respectively, remaining below most commercial limits for herbal drug products. These findings demonstrate that hemp flowers can be safely produced on metal-contaminated soils, reinforcing the suitability of hemp as a robust and versatile crop for the phytomanagement of legacy metal pollution.

NOVELTY STATEMENT

This study assesses the valorization potential of hemp flowers grown on a soil presenting elevated levels of cadmium, lead and zinc. Our findings reveal that flowers could potentially be valorized in phytomanagement contexts, and that they accumulated lead and cadmium less than other plant tissues. This is the first time that such observation is made for lead, while the limited existing data for cadmium in that regard is hereby consolidated.

KEYWORDS

Cannabidiol; *Cannabis sativa* L.; hemp flower; industrial hemp; phytomanagement; trace metals

Introduction

Chemical contamination is a major driver of the global degradation of arable lands (Hossain et al. 2020; Prăvălie 2021). Trace metals and metalloids are particularly problematic in this context, as these potentially toxic elements are non-biodegradable and relatively available to crop uptake (Hossain et al. 2020; Hou et al. 2020). In Europe, it is estimated that 6.2% of agricultural lands exceed risk thresholds for metal(loid) concentrations, and that another 52% require further assessment to ensure safe cultivation (Tóth et al. 2016). Anthropogenic contamination of productive lands with these elements is mostly diffuse, resulting from impure soil amendments or industry and traffic emissions (Nicholson et al. 2003; Hou et al. 2020). This constitutes a severe limitation for soil remediation, as established technologies such as *ex situ* soil washing entail considerable costs, limiting their applicability to pollution hotspots only (Tack and Bardos 2020).

Over the last 20 years, phytomanagement has emerged as a promising economical and sustainable approach for tackling diffuse metal(loid) contamination. This technique aims at the site-tailored mitigation of environmental risks in the most cost-effective manner, through the production of valuable biomass in particular (Robinson et al. 2009; Burges et al. 2018; Saran et al. 2024).

In this context, industrial hemp (*Cannabis sativa* L.) presents interesting properties. Besides low requirements in water, nutrients and pesticide (Amaducci et al. 2015; Michels et al. 2025), the plant demonstrates large biomass yields that are little affected by moderate to high metal(loid) levels (Angelova et al. 2004; Luyckx et al. 2022; De Vos et al. 2023a). Furthermore, the different plant parts are processed into a variety of established applications such as textile, insulation, feed or energy (Crini et al. 2020; Golia et al. 2023; Michels et al. 2025). This versatility is an asset when it comes to phytomanagement, as different end products can be targeted accordingly to the contamination grade of a specific site.

Plant uptake of trace metals depends on a variety of factors. Plants modulate metal accumulation according to their

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physiological status, which is influenced by growth conditions and environmental constraints (Clemens 2001; Uchimiya et al. 2020). Genetic variations within a given plant species also play a part, as different cultivars grown under equal conditions may present uneven metal concentrations (De Vos et al. 2022). Finally, plant uptake is strongly influenced by metal availability, which depends on biological, chemical and physical soil properties, along with the nature and age of the contamination (Robinson et al. 2009; Uchimiya et al. 2020). As a result, extensive ranges of metal concentrations have been reported in hemp. A review conducted by Golia et al. (2023) indicated variations of four orders of magnitude in the above-ground biomass for either Cd, Pb or Zn. Even when bioconcentration is expressed relatively to the total concentration in the substrate (bioconcentration factor, BCF; Shi et al. 2012; Ferrarini et al. 2021; Ofori-Agyemang et al. 2026), reported values remain highly variable, ranging from 0.08 to 31 for Cd, 0.03 to 166 for Pb, and 0.03 to 37 for Zn (Golia et al. 2023). Consequently, the capability of hemp to remediate soils contaminated with these metals remains debated. Some studies have deemed hemp a good candidate for the phytoextraction of Cd (Irshad et al. 2015; Testa et al. 2023; Zhao et al. 2025), Pb (Testa et al. 2023), Zn (Irshad et al. 2015), or the three elements (Angelova et al. 2004; Golia et al. 2023). In contrast, other authors have reported negligible extraction efficiencies, concluding that hemp is not suited for the remediation of soils contaminated with Cd and Pb (Meers et al. 2005) or with the three metals (De Vos et al. 2022; Flajšman et al. 2023).

Different value chains have been promisingly explored for hemp biomass in phytomanagement contexts. De Vos et al. (2023a, 2023b) demonstrated that hemp fibers could be safely manufactured into clothing, as metal concentrations were far below legal standards, including after demineralization and bleaching treatments. Todde et al. (2022) highlighted the potential for bioenergy production through anaerobic digestion or incineration of the whole plant. More recently, several authors have also supported the use of contaminated hemp biomass for biofuel production (Testa et al. 2023; Peroni et al. 2025; Ofori-Agyemang et al. 2026). In contrast, seed production on contaminated sites has often been dismissed, as metal concentration would exceed risk thresholds for food (Linger et al. 2002; Flajšman et al. 2023).

While studies have focused on seeds and fibers in particular, data regarding hemp flowers is essentially lacking. Recently, Sugier et al. (2025) hypothesized that hemp infructescences could be valorized in phytomanagement contexts, but information on flower uptake of Cd, Pb or Zn remains scarce. This contrasts with the increasing attention that this plant part has received as a source of bioactive compounds, the most notable being the non-psychoactive cannabidiol (CBD) (Mark et al. 2020; Kaur and Kander 2023). Cannabidiol is clinically effective in supportive therapy against epilepsy in children, and preclinical studies highlighted that its anti-inflammatory properties can be beneficial for cardiovascular, hepatic and neurogenerative diseases (Andre et al. 2016; Pacher et al. 2020). Consequently, the production of hemp flowers has undergone a steep growth over the last decade. In the United States of America, CBD products were the primary driver for hemp cropping to reach 36,000 ha in 2018 (Mark et al. 2020). This is close to the acreage in Canada, the second hemp producer worldwide after China (Kaur and Kander 2023), where it has been estimated that 29% of the 40,000 cultivated hectares was dedicated to CBD production (Mark et al. 2020).

Cannabinoids are secondary plant metabolites, compounds known to participate in plant defenses against biotic or abiotic stresses. Therefore, it has been hypothesized that physiological stress could promote higher CBD contents (Andre et al. 2016; Husain et al. 2019; Sugier et al. 2025). However, whether metal contamination could lead to such overexpression remains unclear. Recently, Vasilou et al. (2024) did observe significant increases in CBD levels upon soil enrichment with copper. Contrastingly, Yin et al. (2022) did not witness differences among CBD content in flowers after a 15 day exposure to high Cd concentrations. Likewise, Marabesi et al. (2023) exposed hemp plants to Cd doses up to 10 mg L^{-1} in hydroponics, but did not observe larger cannabinoid concentrations.

In this work, hemp was cultivated in the greenhouse on a soil contaminated with Cd, Pb and Zn due to former smelter activities. The objectives were 1) to investigate the influence of metal availability on plant uptake, 2) to assess whether hemp cultivation could help achieve soil remediation, and 3) to evaluate flower valorization potential based on yield, metal concentration and cannabinoid production.

Material and methods

Soil collection

The contaminated soil was sampled 0.5 km away from the former Metaleurop Nord smelter in Noyelles-Godault, France (50°26'15.5"N; 3°00'53.9"E). Historical smelter activities in the area led to a significant enrichment of the topsoil in several trace elements, especially in Cd, Pb and Zn (Sterckeman et al. 2002). A reference soil was sampled 7.5 km away from the smelter (50°29'11.8"N; 3°04'04.1"E). Both locations have been planted with poplar trees for at least 20 years. Both soils are defined as loessic Brunisols in the French soil classification system (Sterckeman et al. 2007). The topsoil layer was removed before soil was collected with a shovel between 5 and 30 cm deep, over 3 randomized locations distant by 5 m. The air-dried soil was manually crushed and homogenized prior to the experiment. Soil properties are provided in Table 1.

Soil characterization

Prior to analysis, soil was crushed with mortar and pestle in the lab and sieved over a 2 mm mesh. All analyses were performed on air-dried soil. For correcting the results expressed on dry weight basis, water content of the air-dried soil was determined after constant weight was achieved by oven-drying at 105 °C.

Soil texture was determined through the hydrometer method as applied by Gavlak et al. (2005), after digestion of the organic matter with 30% (w/w) H_2O_2 at 80 °C. Soil pH-H_2O and pH-KCl were measured with a pH meter (Model 520, Metrohm, Switzerland) in a 1:5 (w/w) soil:solution ratio equilibrated overnight, respectively with deionized

Table 1. Physical and chemical properties of the soils used for the experiment (mean ± standard deviation, $n = 4$).

	Contaminated soil	Reference soil
pH-H_2O	8.2 ± 0.2	6.5 ± 0.1
pH-KCl	7.1 ± 0.1	5.4 ± 0.1
Sand (%)	27.6	43.1
Silt (%)	53.5	40.4
Clay (%)	18.9	16.5
Organic carbon (%)	1.98 ± 0.12	2.23 ± 0.06
Inorganic carbon (%)	0.05 ± 0.02	< 0.005
Field capacity (%)	31.1 ± 0.4	33.2 ± 0.3
Fe (%)	1.71 ± 0.03	1.28 ± 0.01
Al (%)	1.22 ± 0.04	0.94 ± 0.02
Total N (mg kg^{-1})	1570 ± 10	1800 ± 30
Inorganic N (mg kg^{-1})	7.6 ± 1.8	9.2 ± 0.9
P (mg kg^{-1})	596 ± 12	376 ± 17
Ca (mg kg^{-1})	5198 ± 143	2741 ± 72
Mg (mg kg^{-1})	2468 ± 44	1818 ± 17
Na (mg kg^{-1})	220 ± 25	215 ± 39
K (mg kg^{-1})	1376 ± 261	830 ± 245
Cd (mg kg^{-1})	13.0 ± 0.1	1.08 ± 0.04
Pb (mg kg^{-1})	664 ± 10	45.0 ± 1.7
Zn (mg kg^{-1})	1048 ± 13	100 ± 3
$CaCl_2$-extractable Cd (mg kg^{-1})	0.46 ± 0.12	0.15 ± 0.05
$CaCl_2$-extractable Pb (mg kg^{-1})	0.32 ± 0.21	0.031 ± 0.022
$CaCl_2$-extractable Zn (mg kg^{-1})	2.5 ± 1.0	2.6 ± 0.8

water and KCl 1 *M*. Organic carbon, inorganic carbon and total nitrogen contents were determined with a CN analyzer (Primacs SNC-100, Skalar, the Netherlands). Field capacity was assessed by weight difference from the amount of water retained in pots 72 h after saturating the soil with water. Pseudo-total concentrations were determined after digestion with *aqua regia* (1:3 mixture of HNO_3 and HCl) (Ure 1990). Inorganic nitrogen was determined as the sum of nitrate and ammonium nitrogen, respectively analyzed according to ISO methods 13395:1996 and 11732:1997 with a continuous flow auto-analyzer (Chemlab System 4, Skalar, The Netherlands) in a 1:4 KCl 1 M extract. $CaCl_2$-extractable concentrations of Cd, Pb and Zn were determined by extracting 4 g of soil with 50 ml of 0.01 M $CaCl_2$ for 2 h before filtration over a filter paper (pore size 12 µm) (Van Ranst et al. 1999).

Greenhouse experiment

Pots (average radius 9.7 cm, height 22 cm) were filled with 5.5 kg air-dried soil. Throughout the experiment, watering was performed with rainwater 3 times per week to maintain 60% of the field capacity. No drainage was observed during watering. Finely ground soluble mineral fertilizer (proportions in N, P_2O_5, K_2O and SO_3 of 16, 9, 20 and 30%, Compo, Germany) was evenly spread over the pot surface at two moments: an initial dose equivalent to 96 kg N ha^{-1} one week prior to planting, and a supplementary dose equivalent to 32 kg N ha^{-1} after 43 days.

Seeds of *Cannabis sativa L.* subsp. Fedora 17 (Pevgrow, Spain) were germinated on a wetted tissue in a LDPE box with perforated cover under ambient light and indoor temperature conditions. Once seedlings were approximately 2 cm long, four germinated seeds were planted into the pots. After 15 days, one viable plant per pot was selected and the others were discarded. A minimal temperature of 15 °C was kept in the greenhouse at all time, with average day and night temperatures (± standard deviation) of respectively 21.5 ± 2.5 °C and 18.7 ± 1.7 °C. The average relative humidity level was 60.3 ± 9.4%. A minimal photoperiod of 16 h day^{-1} was ensured during the vegetative growth phase, which was adjusted to 10 h day^{-1} after 55 days to stimulate flowering. Four replicates were used for the contaminated soil and 7 for the reference soil, a difference due to more pots being originally prepared as back-up for this soil and finally considered throughout in the experiment.

Soil pore water was sampled at four intervals (cultivation days 14, 38, 59, 79) using Rhizon soil moisture samplers (pore size 0.15 µm, Eijkelkamp, the Netherlands). Samplers were inserted into diagonal 45° holes previously drilled with a wooden spike, after which surrounding soil was gently compressed to ensure adequate sealing. One drop of concentrated HNO_3 was added to the pore water samples, which were stored at 8 °C prior to analysis. Pore water collection was not always achievable for all pots. The number of replicates for these measurements varied between 1 to 3 for the reference soil, and 3 to 4 for the contaminated soil.

Plants were harvested after 79 days. Flower buds and leaves were cut off the stem. The stem was separated into long bast fibers and shives by manual peeling. Roots were manually separated from the soil, thoroughly rinsed with deionized water and soaked for 20 min in a 20 mmol L^{-1} EDTA solution. Flowers were dried upside down in a pre-heated oven at 28 °C for 6 days, subsequently milled in a ball mill and stored away from light at 8 °C, according to standard practice for the preservation of cannabinoids (Al Ubeed et al. 2022). A sub-sample was then further dried at 105 °C until constant weight, indicating that the residual moisture content in flowers was 11.9 ± 0.8%, which corresponds well to commercial standards (Al Ubeed et al. 2022). All other plant parts were dried at 65 °C for 48 h, milled in a ball mill and kept at room temperature. For elemental analysis, about 0.2 g of plant tissue was sonicated for 30 min in pico-pure grade HNO_3, then digested in the same matrix for 20 min at 180 °C and 1000 W in a dedicated microwave (Mars 6, CEM, USA).

Trace metal analysis

Elemental concentrations were analyzed through Inductively Coupled Plasma-Optical Emission Spectrometry (ICP-OES, iCAP 7400DUO, Thermo Fisher Scientific, USA), or Inductively Coupled Plasma-Mass Spectrometry (ICP-MS, NexION 350D, PerkinElmer, USA) when limit of quantification was too low for the former. Quality control was performed using two certified reference materials systematically digested alongside samples: SCP-140-025-001 (SCP Science, Canada) and CRM020-50G (Merck, Germany) for soil samples, and SRM1547 and SRM1570a (Merck, Germany) for plant samples. All recovery rates for reference materials ranged between 92 and 113%.

Cannabinoid analysis

Approximately 0.5 g of dried flower were extracted with 25 ml of dichloromethane during 30 min in a sonication bath, and extracts were filtered through a 0.22 µm PTFE filter before analysis. This analysis was conducted on different replicates than metal analysis, as some plants did not produce enough flower material. Thus, cannabinoid analysis was performed on 6 replicates for each series (reference and contaminated), out of which 4 replicates were grown under the same conditions except the soils were amended with 1% w/w spruce pine biochar (Greenpoch, Belgium).

Cannabidiol and delta-9-transtetrahydrocannabinol (Δ9-THC) were analyzed using Gas Chromatography Mass Spectrometry (GC-MS) using an Agilent 7890N system (Agilent, USA). The instrument was equipped with a capillary column (Agilent, VF-5ms, 40m length, 0.25mm internal diameter) impregnated with an 0.25μm phenyl-methylsiloxane film.

Statistical analysis

All analyses were performed using the R Statistical Software (v4.2.3, R Core Team 2023). Equality of variance between sample groups was assessed with Levene's test. Statistical difference between groups were assessed through Wilcoxon's rank-sum test when sample variance was unequal, or through Student's t-test otherwise. Tests were considered significant at $p \leq 0.05$.

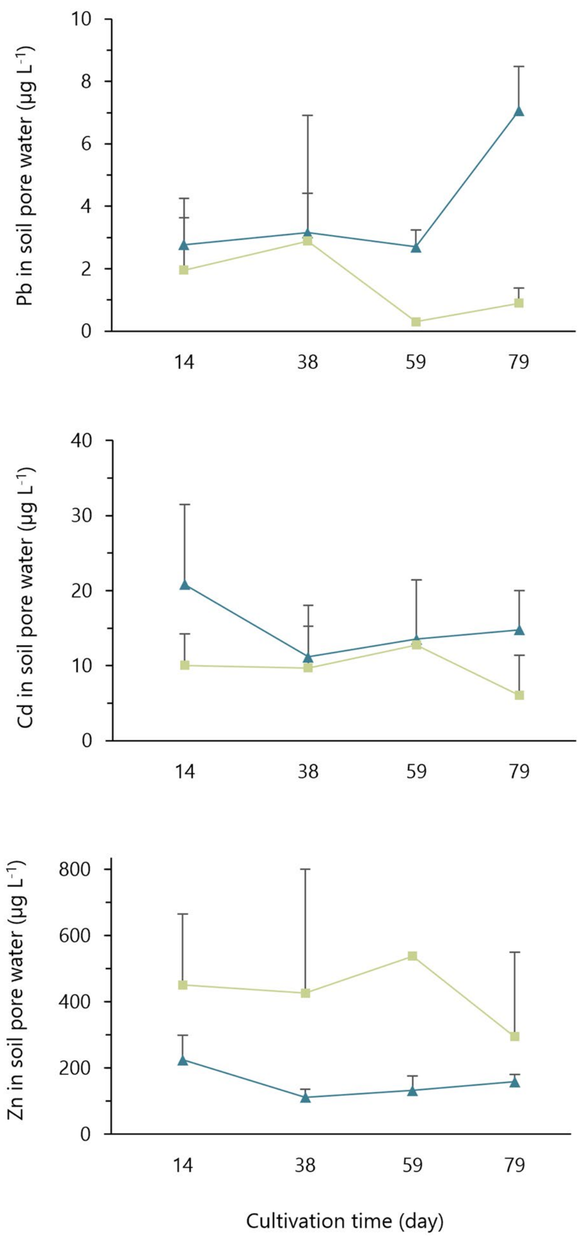

Figure 1. Metal concentration in soil pore water (µg L^{-1}) during cultivation at day 14, 38, 59 and 79 (mean with standard deviation as error bars, n = between 3 and 4 for the contaminated soil, between 1 and 3 for the reference soil). Blue triangle plots (—▲—) are for the contaminated soil, green rectangular plots (—■—) are for the reference soil.

Results

Soil contamination

Pseudo-total trace metal concentrations in the contaminated soil were 15 to 30 times higher than regional baseline values for the surface horizon as reported by Sterckeman et al. (2007; 0.44 mg Cd kg^{-1}, 36 mg Pb kg^{-1} and 70 mg Zn kg^{-1}). They were also 1.3 to 4.2 times above the risk values of the Finnish legislation (10 mg Cd kg^{-1}, 200 mg Pb kg^{-1} and 250 mg Zn kg^{-1}), a regulatory framework that is deemed a good approximation for European standards (Tóth et al. 2016). In contrast, metal concentrations in the reference soil were close to regional background levels, although moderate enrichments were still observed for Zn (1.4-fold) and Cd (2.5-fold). For comparison, aluminum and iron contents roughly corresponded to the 5th and 25th percentile of regional baseline values respectively (Sterckeman et al. 2007).

Calcium chloride extracts differed less between soils than pseudo-total concentrations. Although Pb remained an order of magnitude higher in the contaminated soil, contrasts were less pronounced for Cd and Zn. Differences between the two soils were even further attenuated in the pore water (Figure 1): soluble Pb and Cd concentrations were comparable in both soils, while Zn levels in the pore water were even higher in the reference soil despite the lower pseudo-total content.

The pH was markedly higher in the contaminated soil. This can be attributed to prior liming, as white chalk-like particles were observed and found to contain 38% carbonates (based on loss on ignition between 550 °C and 950 °C), 27% calcium and 0.2% magnesium by weight. Although the exact timing of the liming event is unknown, it is likely anterior to the establishment of poplar plantations on the site over 20 years ago. Because the reference soil was intended to represent optimal conditions for cultivation, and since suitable pH values for hemp range from 5.8 to 7.5 (Amaducci et al. 2015), the original pH of the reference soil was left unmodified for the experiment.

Plant yield

Yields of total above ground biomass (AGB), flowers and stems are presented in Table 2. Importantly, flower yield was not affected by metal contamination. Although stem yield was 1.5 times lower in the contaminated soil, the difference was not statistically significant.

Table 2. Yield per plant for total above ground biomass (AGB), flowers and stems (g dry weight, mean ± standard deviation, $n = 7$ for the reference soil, $n = 4$ for the contaminated soil).

	Total AGB	Flower	Stem
Contaminated soil	13.8 ± 8.5*a*	1.4 ± 1.2*a*	5.8 ± 4.0*a*
Reference soil	17.9 ± 6.3*a*	1.4 ± 0.5*a*	9.0 ± 4.1*a*

For a given plant part, different letters indicate significant differences according to a one-tailed Student t-test ($p \leq 0.05$).

Extrapolating the observed biomass to field scale by using a density of 70 plants m^{-2} (Struik et al. 2000; Amaducci et al. 2015), the contaminated soil would produce an estimated 9.7 ton ha^{-1} of AGB. This value is comparable to the lower range of reported yields for the same cultivar grown on uncontaminated soils (Struik et al. 2000).

Plant concentration and metal uptake

Metal concentrations in the different plant tissues are detailed in Figure 2. The three elements exhibited various distribution patterns over the plant parts. Lead concentrations showed a clear concentration gradient in the order root>

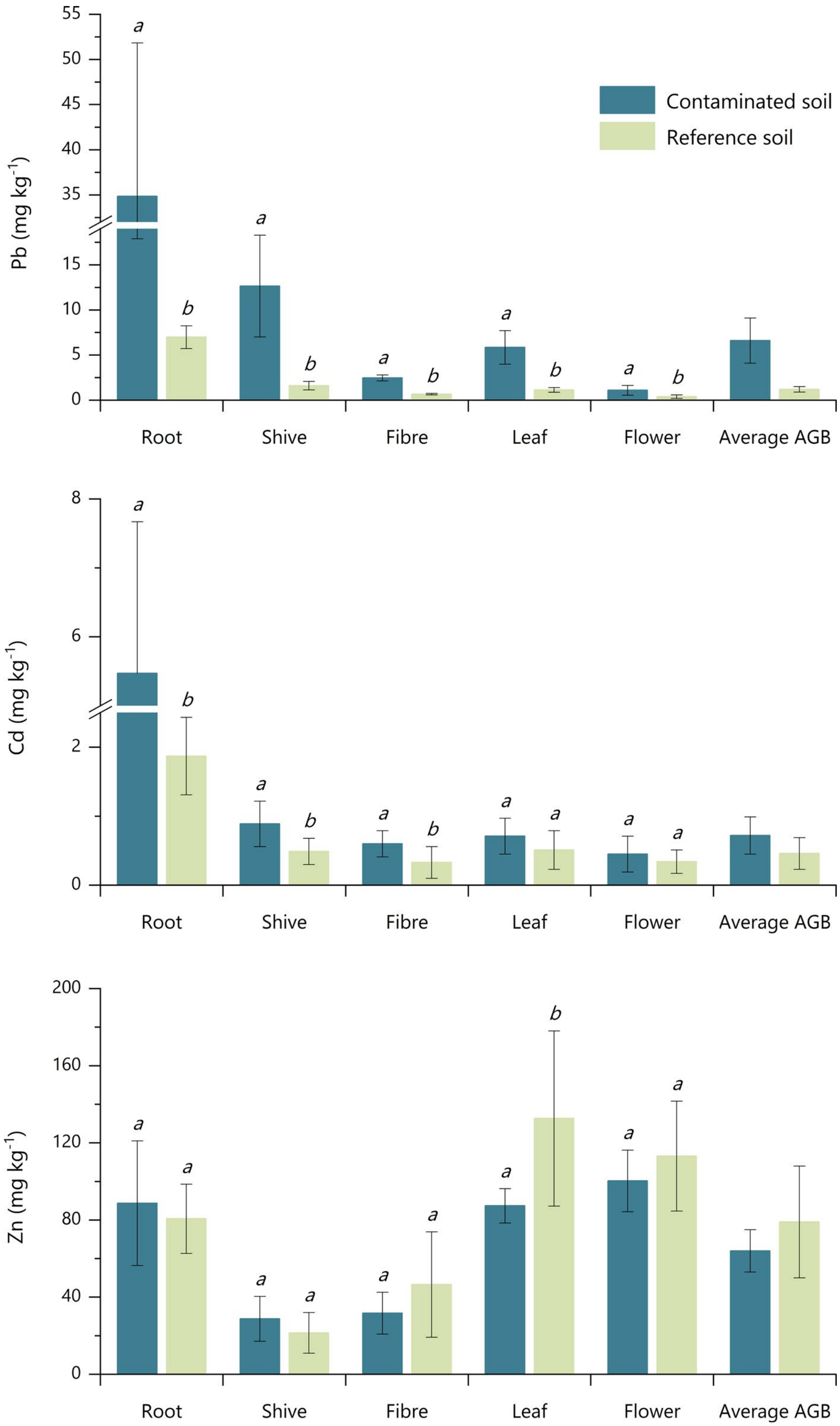

Figure 2. Metal concentrations in the different plant parts (mg kg^{-1}dry weight, mean with standard deviation as error bar) and weighted concentration in the above ground biomass (AGB, mg kg^{-1}dry weight, weighted mean with weighted standard deviation as error bar) ($n=7$ for the reference soil, $n=4$ for the contaminated soil). Blue bars are for the contaminated soil, green bars for the reference soil. For a given plant part and a given element, different letters indicate significant differences according to the Wilcoxon rank-sum test ($p \leq 0.05$).

Table 3. Plant uptake as a function of the pseudo-total, $CaCl_2$-extractable and soluble fractions in the soil.

	Zn		Pb		Cd	
	Contaminated soil	Reference soil	Contaminated soil	Reference soil	Contaminated soil	Reference soil
BCF	0.061	0.79	0.010	0.026	0.055	0.43
Uptake of pseudo-total fraction (%)	0.015	0.257	0.003	0.009	0.014	0.139
Uptake of $CaCl_2$-extractable fraction (%)	6.4	9.9	5.1	12.6	0.4	1
Uptake of soluble fraction (%)	548	323	2276	1395	64	84

Bioconcentration factor (BCF) was calculated as the weighted above-ground biomass concentration divided by the pseudo-total soil concentration. Plant uptake of the pseudo-total, $CaCl_2$-extractable and soluble fractions were calculated as the average metal quantity in one plant divided by the amount present in these fractions in one pot. The soluble fraction was calculated as the pore water concentration averaged over the four sampling times.

shive > leaf > fiber > flower. A similar trend was observed for Cd, even though differences among above ground parts were less obvious. For these two elements, flowers exhibited the lowest concentrations, of 0.45 and 1.1 mg kg^{-1} on the contaminated soil for Cd and Pb respectively. Zinc seemed as easily taken up in some above ground plant parts than in roots, with concentrations in the following order: leaf ~ flower > root > fiber ~ shive.

The influence of soil contamination did not always reflect in the plant, and bioaccumulation patterns were distinct according to the element. Lead concentrations were all significantly higher for the contaminated soil, with the highest difference observed in shives (factor 7.8) and the lowest in flowers (factor 2.8). The contrast was less marked for Cd, with concentrations differing by a factor 1.9 on average, and significant differences only found in fibers, shives and roots. Similarly to Pb, the smallest difference was observed for flowers (factor 1.3). This further suggests that flowers accumulate these two metals the least among all plant tissues. Zinc concentrations were similar regardless of the soil, and occasionally even higher for the reference (1.5 times significantly higher in leaves for example). Fluctuations among bioaccumulation patterns were also visible in BCF values (Table 3), which suggested a preferential uptake in the order Zn > Cd > Pb. Bioaccumulation factors were particularly low for all 3 elements in the contaminated soil (0.01-0.06).

Over the course of the experiment, plants extracted only a negligible fraction of the pseudo-total metal content in a given pot, especially for the contaminated soil. Plant uptake of the $CaCl_2$-extractable quantity was relatively limited too, with a maximum of 12.6% for Pb in the reference soil. Contrastingly, the metal pool in the soil solution was taken up an equivalent of 0.6 to 22.8 times, being depleted several times for Zn and Pb.

Cannabinoid content in flowers

Metal contamination did not stimulate the production of cannabinoids, as concentrations were very similar regardless of the soil type (Table 4). Cannabidiol concentrations were close to previously reported contents for cultivar Fedora 17, which usually range between 1.5 and 2.0% (Husain et al. 2019; Glivar et al. 2020). Delta-9-THC concentrations were two orders of magnitude lower than the European threshold for industrial hemp of 0.3% (European Parliament and Council 2021), which is not uncommon for this low-THC cultivar (Glivar et al. 2020).

Table 4. Cannabinoid concentration in dried hemp flowers (% w/w, mean ± standard deviation, $n = 6$).

	CBD	Δ9-THC
Contaminated soil	1.05 ± 0.13*a*	0.0025 ± 0.0005*a*
Reference soil	1.08 ± 0.45*a*	0.0027 ± 0.0010*a*

For a given cannabinoid, different letters indicate significant differences according to a two-tailed Student t-test ($p \leq 0.05$).

Discussion

Metal availability and plant uptake

The difference in total metal content between soils was not proportionally reflected in plant tissues, likely because of a lower metal availability in the contaminated soil. Indeed, the $CaCl_2$-extractable and the soluble concentrations showed smaller differences between soils and correlated more with plant uptake than the pseudo-total content. It is well established that the phytoavailability of Cd, Pb and Zn in soils decreases with higher pH, as acidic conditions promote more soluble and exchangeable forms that are readily taken up by plants (Hou et al. 2020; Uchimiya et al. 2020). Thus, the substantially higher pH in the contaminated soil likely played an important role in reducing the phytoavailability of Cd and Pb, hence plant concentrations. The long-term presence of the contamination may further explain the limited phytoavailability, as suggested previously for the same area (Ofori-Agyemang et al. 2024). Through aging processes, legacy contamination becomes progressively less bioavailable as metals increasingly associate with more stable soil mineral phases (Smolders et al. 2009).

Plant metal concentrations in our experiment fall within the lower range of values previously reported for hemp, as summarized in the recent review conducted by Golia et al. (2023). However, the highest concentrations in the literature correspond to studies conducted under hydroponic conditions or on soils spiked with soluble metal salts, both of which greatly increase metal availability. For example, Petrová et al. (2012) observed stem concentrations that were 850-fold higher for Zn and up to nearly 100,000-fold higher for Cd compared to our findings when hemp was grown hydroponically in solutions containing 5,000 μmol L^{-1} of metals. This concentration is approximately 2,000 (Zn) to 250,000 (Pb) times greater than pore water concentrations in our study, explaining the substantial differences in plant uptake. Similarly, Zhao et al. (2022) spiked a soil with soluble metal nitrate salts to reach levels of 40 mg Cd kg^{-1} and 1,500 mg Pb kg^{-1}, resulting in AGB concentrations 98 and 81 times greater than these observed here. Shi et al. (2012)

reported a comparable effect: the addition of 25 mg kg^{-1} of soluble $CdCl_2$ to the soil resulted in Cd shoot concentrations 16 to 46 times higher than the values reported here. These comparisons highlight the strong influence of metal speciation and availability on plant uptake, and underscore why field-aged contamination such as in our soil typically leads to much lower accumulation than freshly spiked or hydroponic systems.

In fact, plant concentrations in the present study are comparable to those measured in aged contaminated soils. Flajšman et al. (2023) reported AGB Zn and Cd concentrations that were respectively 1.4 and 1.2 times lower than ours under equivalent soil contamination (1.5 and 1.9 times lower). Although metal availability was not directly assessed, the similar soil acidity (pH-$CaCl_2$ = 7.0) can be linked with the analogous plant uptake. Angelova et al. (2004) also reported comparable Zn and Cd plant concentrations in a field trial where smelter activities had resulted in soil contents of respectively 536 and 12.2 mg kg^{-1}. Soil pH was also similar (pH-H_2O = 7.7). Their study documented a substantially higher Pb accumulation, with shoot concentrations reaching up to 45 mg kg^{-1} despite a soil content of only 200 mg kg^{-1}. However, this higher uptake was likely due to ongoing pollution during their field experiment, which may have led to larger metal availability or direct contamination of the above ground plant parts.

Conversely, Canu et al. (2022) measured plant concentrations that were comparable for Pb but approximately 6 times higher for Zn and Cd on a soil presenting similar total metal contents. Their results indicated relative availability based on the DTPA extraction test of 20, 26 and 64% for Zn, Pb and Cd respectively. These values are particularly high for the 3 metals (Kashem et al. 2007), which was likely influenced by their more acidic soil conditions (pH-H_2O = 6.4). This lower pH may also explain the higher Zn and Cd uptake that they observed compared with our results. More recently, Jaros-Tsoj et al. (2025) reported hemp leaf concentrations that were roughly equivalent to ours (differing by a factor of only 0.99-1.66), even though total soil concentrations were substantially larger (51 mg Cd kg^{-1}, 2,940 mg Pb kg^{-1} and 8,057 mg Zn kg^{-1}). Their $CaCl_2$-extractable fractions were very similar (0.3 mg Pb kg^{-1}, 2.9 mg Zn kg^{-1}) or even lower (0.2 mg Cd kg^{-1}) than those measured in our soil, illustrating that low bioavailability can mitigate the influence of high total metal loads on plant uptake.

Plant concentrations measured hereby also align with findings in the same study area. Ofori-Agyemang et al. (2024) tested different soil amendments on a soil with similar metal concentrations and pH (pH-H_2O = 7.9), sampled 700 m from the same smelter. In their unamended soil, shoot concentrations of Zn (59 mg kg^{-1}) and Pb (8.5 mg kg^{-1}) closely matched our results while their Cd concentration (2.1 mg kg^{-1}) was approximately 2.9 times higher. This difference corresponds with the higher pore water Cd concentration that they measured (60 µg L^{-1} versus 15 µg L^{-1} in our study). Similarly, De Vos et al. (2022) evaluated six hemp varieties in a soil sampled only 200 m from our site and reported plant concentrations within the same order of magnitude (0.55-1.2 µg Cd g^{-1}, 5.5–14 µg Pb g^{-1} and 15–33 µg Zn g^{-1}).

Overall, studies comparing hemp varieties under similar conditions typically show variations of less than a factor two (Zhao et al. 2022; Testa et al. 2023) or three (Shi et al. 2012; De Vos et al. 2022; Flajšman et al. 2023) within a given plant part for Cd, Pb or Zn. This suggests that cultivar choice plays only a limited role in explaining the substantial discrepancies in reported hemp metal concentrations. Instead, metal availability, as determined by site-specific pH, contamination history and metal speciation, appears to be the dominant factor.

Our data suggests a preferential uptake pattern in the order root > shive > leaf > fiber > flower for Pb and Cd. This aligns with the existing literature, where similar gradients root > AGB (Petrová et al. 2012; Shi et al. 2012; Ahmad et al. 2015; Irshad et al. 2015; Testa et al. 2023), leaf > fiber (Linger et al. 2002; Angelova et al. 2004) or leaf > stem (Ahmad et al. 2015; Canu et al. 2022; Luyckx et al. 2022; Testa et al. 2023) are often reported for these metals.

Remarkably, our results suggest that flowers accumulate the least Pb and Cd among plant tissues. This is consistent with Marabesi et al. (2023), who observed an overall root ≫ leaf > stem > flower pattern for Cd in hydroponically grown hemp. In contrast, Angelova et al. (2004) found that flowers were the most accumulating tissue for Cd and Pb. However, this discrepancy may be attributed to ongoing smelter activities during their field trial, which likely resulted in additional metal inputs to the upper plant parts. In general, lower concentrations in flower align with the broader biological principle that plants tend to exclude potentially toxic elements from reproductive organs. Similar trends have been observed for Cd and Pb in other plant species (Hladun et al. 2015; Perlein et al. 2021), as well as for chromium in hemp (Ferrarini et al. 2021). This exclusion mechanism also extends to hemp seeds, which tend to accumulate less Cd or Pb than other hemp tissues (Linger et al. 2002; Canu et al. 2022; Testa et al. 2023; Zhao et al. 2025).

Remediation potential

Our data suggests a low extraction potential for the three studied elements. Bioconcentration factors were much lower than 1 for the contaminated soil, indicating a tendency to prevent metal translocation to above ground tissues. In similar fashion as for plant concentrations, BCF values were among the lower range previously reported for hemp (Golia et al. 2023), but of the same magnitude as in studies assessing field-aged contaminated soils (Angelova et al. 2004; Canu et al. 2022; Flajšman et al. 2023; De Vos et al. 2023a; Ofori-Agyemang et al. 2024; Jaros-Tsoj et al. 2025).

Biomass production was not significantly affected by soil contamination in our experiment, whether considering total above-ground biomass, stem yield or flower yield. This is particularly noteworthy given that the pH of the contaminated soil exceeded the optimal range for hemp cultivation of 5.8 – 7.5 (Amaducci et al. 2015), whereas the reference soil fell into that range. Hemp robustness against Cd, Pb and Zn toxicity was previously reported, with unaffected yields between reference and contaminated conditions under similar metal contents in the soil (Angelova et al. 2004; Luyckx et al. 2022). Nonetheless, caution is warranted, as De Vos et al. (2023a) observed a lower stem yield working in the neighborhood of our sampling location compared with non-contaminated conditions.

In our case, it is likely that the elevated pH in the contaminated soil lowered metal availability, hence toxicity, as previously observed in other plants for Cd (De Oliveira et al. 2016) or Pb (Dayton et al. 2006). $CaCl_2$-extractable fractions can help predict metal toxicity (Uchimiya et al. 2020), and were proportionally lower in the contaminated soil than in the reference soil compared with their respective total content. Thus, even though the pH of our contaminated soil was not ideal for hemp cultivation, it may have alleviated metal toxicity and supported retaining unaltered yields.

Based on weighted AGB concentrations, metal removal rates can be calculated of respectively 0.15, 0.015 and 0.0017 mg $kg^{-1}year^{-1}$ for Zn, Pb and Cd, assuming an AGB yield of 9.7 ton ha^{-1} as extrapolated in section 3.2, a soil water content equal to 60% of the field capacity, a soil bulk density of 1.4 ton m^{-3} and a treated depth of 30 cm. These rates suggest that achieving soil remediation to meet legal thresholds based on total soil content, such as specified in Tóth et al. (2016), would require thousands of years. Therefore, phytomanagement should rather be recommended, with an emphasis on biomass valorization and a concomitant gradual attenuation of the most labile metal fractions over time. Assuming no replenishment of the $CaCl_2$-exchangeable pool, removing an amount equivalent to this fraction would still require 223 years for Cd, but only 17 years for Pb and 14 years for Zn. Even more encouraging in that perspective are the findings of Ofori-Agyemang et al. (2026), who highlighted a remarkable 79–96% decrease in the 0.01 M $Ca(NO_3)_2$-extractable fractions for Cd, Pb and Zn after only two years of hemp cultivation, at a site located 250 m away from our sampling location. Notably, their $Ca(NO_3)_2$-extractable concentrations were approximately 2 times (Cd, Zn) to 15 times (Pb) higher than our $CaCl_2$-extractable fractions. In any case, additional data on release kinetics would be valuable to confirm the long-term attenuation potential of labile metal pools, as these tend to replenish from larger immobile fractions (Robinson et al. 2009).

Hemp flower valorization

Our findings suggest diverse valorization pathways for the different plant parts. Fiber concentrations were at least 36 times lower than the international Oeko-Tex® standard 100 label for textile fibers of 40 and 90 mg kg^{-1} for Cd and Pb (Oeko-Tex 2025). Furthermore, anaerobic digestion of the whole plant to produce biogas would be an option too, with AGB concentrations under the corresponding permitted levels in Flanders of 1500, 300 and 6 mg kg^{-1} for Zn, Pb and Cd respectively, considering that digestate would subsequently be used as bio-based fertilizer (Flemish Government 2012). The digestate could be theoretically applied in France too, given the corresponding French limits of 600 mg Zn kg^{-1}, 180 mg Pb kg^{-1} and 3 mg Cd kg^{-1} (French standardization association 2006), assuming a composting step to allow recognition as organic fertilizer (Trémier et al. 2013).

Among all plant parts, flowers exhibited the lowest Pb and Cd concentrations. Notably, these levels are below the legal safety thresholds for herbal drugs and products in the European Union, respectively 5 and 1 mg kg^{-1} (Council of Europe 2017), by a factor 2.2 at least. Since Zn levels are not regulated, flowers from our experiment could be marketed in Europe without restriction. Compliance with other regional standards may vary, as regulatory thresholds for herbal medicines range from 0.3 to 5 mg kg^{-1} for Cd and between 0.5 to 10 mg kg^{-1} for Pb. Nonetheless, our concentrations still comply with 7 out of 9 major pharmacopoeias worldwide (Inada et al. 2023).

These findings open promising opportunities for valorizing hemp flowers cultivated on contaminated soils. This is remarkable given the rising demand for this plant part, which has become an essential product in the hemp market (Mark et al. 2020; Kaur and Kander 2023). The fact that flower yield remained unaffected in our experiment further supports this potential. Additionally, flowers reach maturity at the same growth stage as the fiber and can be harvested as a co-product of fiber hemp (Crini et al. 2020). This could prove beneficial in phytomanagement: fiber can be the primary non-food product, with flowers providing an additional revenue stream on the condition that metal concentrations comply with regional safety standards.

We did not observe any influence of metal contamination on cannabinoid production. This contrasts with Husain et al. (2019), who reported a significant increase in cannabinoid content that the authors attributed to the presence of metals in the soil. However, metal (Cd, Ni, Pb, Hg) and metalloid (As) concentrations in their soil were arguably low, below the background levels for agricultural lands in the region of the present study (Sterckeman et al. 2007). This suggests that other factors than metal contamination may have influenced cannabinoid production, a hypothesis further supported by the absence of significant changes in the expression of metal transporter genes in their experiment.

Other authors reported higher cannabinoid contents under the influence of copper. Vasilou et al. (2024) spiked a clay loam soil with 200 ppm $Cu(NO_3)_2$ and observed an increase in CBD concentration in both leaves and flowers by up to 61%. Similarly, Cahill et al. (2024) observed an increase of 18 to 34% for CBD and THC in hemp inflorescences after foliar spraying of $CuSO_4$ and CuO nanoparticles. Unlike the toxic elements Cd and Pb, copper is an essential plant micronutrient, and only larger amounts would result in physiological stress. In that regard, it can be noted that Cahill et al. observed a significant cannabinoid increase for moderate CuO doses only (50 and 100 ppm) but not at the highest dose (500 ppm).

Our findings are consistent with those of Yin et al. (2022), who investigated the impact of Cd soil contamination on CBD and THC expression. When hemp plants were exposed to a concentration of 100 mg kg^{-1} of Cd in the soil 15 days before harvest, no increase in cannabinoid content was observed in flowers. Similarly, Marabesi et al. (2023) enriched a hydroponic solution with up to 10,000 μg L^{-1} of Cd and did not detect higher CBD or THC levels.

Conclusion

Hemp did not exhibit phytoextraction potential in this experiment, as removal rates remained limited for all three metals. Instead, our findings support phytomanagement strategies that emphasize biomass valorization enabled by the moderate metal accumulation in plant tissues. In particular, the production of hemp flowers on contaminated land appears promising: flower yield was unaffected by soil contamination, and inflorescences accumulated the lowest concentrations of Cd and Pb among all plant parts. Notably, flower concentrations for both elements remained below most commercial thresholds worldwide. The fact that cannabinoid production was unchanged further indicates that the commercial value of hemp flowers would be preserved.

Plant metal concentrations and the measured soil metal availability were consistent with observations from other studies conducted on historically contaminated, non-spiked soils, reinforcing the relevance of our results to moderately polluted field conditions. Collectively, these findings suggest new market opportunities for hemp in a context of phytomanagement, particularly given the growing demand for flower-derived products over the past decade. Furthermore, they bolster the versatility of hemp, as a wider range of products could be targeted according to site contamination level. Since most hemp flower products consists in liquid extracts, future research should investigate the fate of metals during extraction processes.

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Disclosure statement

No potential conflict of interest was reported by the authors.

Data availability statement

Research data detailed in the present article is available at the following repository address: https://doi.org/10.6084/m9.figshare.28776419 (Delemazure 2025).

References

Ahmad A, Hadi F, Ali N. 2015. Effective phytoextraction of cadmium (Cd) with increasing concentration of total phenolics and free proline in *Cannabis sativa* (L) plant under various treatments of fertilizers, plant growth regulators and sodium salt. Int J Phytoremediation. 17(1-6):56–65. doi:10.1080/15226514.2013.828018.

Al Ubeed HM, Wills R, Chandrapala J. 2022. Post-harvest operations to generate high-quality medicinal cannabis products: a systemic review. Molecules. 27(5):1719. doi:10.3390/molecules27051719.

Amaducci S, Scordia D, Liu FH, Zhang Q, Guo H, Testa G, Cosentino SL. 2015. Key cultivation techniques for hemp in Europe and China. Ind Crops Prod. 68:2–16. doi:10.1016/j.indcrop.2014.06.041.

Andre C, Hausman JF, Guerriero G. 2016. *Cannabis sativa*: the plant of the thousand and one molecules. Front Plant Sci. 7:19. doi:10.3389/fpls.2016.00019.

Angelova V, Ivanova R, Delibaltova V, Ivanov K. 2004. Bio-accumulation and distribution of heavy metals in fibre crops (flax, cotton and hemp). Ind Crops Prod. 19(3):197–205. doi:10.1016/j.indcrop.2003.10.001.

Burges A, Alkorta I, Epelde L, Garbisu C. 2018. From phytoremediation of soil contaminants to phytomanagement of ecosystem services in metal contaminated sites. Int J Phytoremediation. 20(4):384–397. doi:10.1080/15226514.2017.1365340.

Cahill MS, Arsenault T, Bui TH, Zuverza-Mena N, Bharadwaj A, Prapayotin-Riveros K, White JC, Dimkpa CO. 2024. Copper stimulation of tetrahydrocannabinol and cannabidiol production in hemp (*Cannabis sativa* L.) is copper-type, dose, and cultivar dependent. J Agric Food Chem. 72(13):6921–6930. doi:10.1021/acs.jafc.3c07819.

Canu M, Mulè P, Spanu E, Fanni S, Marrone A, Carboni G. 2022. Hemp cultivation in soils polluted by Cd, Pb and Zn in the mediterranean area: sites characterization and phytoremediation in real scale settlement. Appl Sci. 12(7):3548. doi:10.3390/app12073548.

Clemens S. 2001. Molecular mechanisms of plant metal tolerance and homeostasis. Planta. 212(4):475–486. doi:10.1007/s004250000458.

Council of Europe. 2017. European pharmacopoeia. 6th ed. European directorate for the Quality of Medicines & Healthcare. (Herbal drugs. 6:1433)

Crini G, Lichtfouse E, Chanet G, Morin-Crini N. 2020. Applications of hemp in textiles, paper industry, insulation and building materials, horticulture, animal nutrition, food and beverages, nutraceuticals, cosmetics and hygiene, medicine, agrochemistry, energy production and environment: a review. Environ Chem Lett. 18(5):1451–1476. doi:10.1007/s10311-020-01029-2.

Dayton EA, Basta NT, Payton ME, Bradham KD, Schroder JL, Lanno RP. 2006. Evaluating the contribution of soil properties to modifying lead phytoavailability and phytotoxicity. Environ Toxicol Chem. 25(3):719–725. doi:10.1897/05-307R.1.

De Oliveira VH, Melo L, Abreu CA, Coscione AR. 2016. Influences of soil pH on cadmium toxicity to eight plant species. Ecotoxicol Environ Contam. 11(1):45–52. doi:10.5132/eec.2016.01.07.

De Vos B, De Souza MF, Michels E, Meers E. 2023a. Industrial hemp (*Cannabis sativa* L.) field cultivation in a phytoattenuation strategy and valorization potential of the fibers for textile production. Environ Sci Pollut Res Int. 30(14):41665–41681. doi:10.1007/s11356-023-25198-z.

De Vos B, De Souza MF, Vermeire S, Vermeir P, Michels E, De Raeve A, Meers E. 2023b. The potential to produce safe textiles in a phytoattenuation strategy: the fate of Cd and Pb during industrial treatments on contaminated hemp fibers. Ind Crops Prod. 204:117408. doi:10.1016/j.indcrop.2023.117408.

De Vos B, Souza MF, Michels E, Meers E. 2022. Industrial hemp (*Cannabis sativa* L.) in a phytoattenuation strategy: remediation potential of a Cd, Pb and Zn contaminated soil and valorization potential of the fibers for textile production. Ind Crops Prod. 178:114592. doi:10.1016/j.indcrop.2022.114592.

Delemazure A. 2025 Jul 01. Soil metal concentrations, plant yield, plant metal concentrations and cannabinoid concentrations in hemp flowers grown on a soil contaminated with cadmium, lead and zinc [dataset] Figshare. [accessed date 2026 Feb 02]. doi:10.6084/m9.figshare.28776419.

European Parliament and Council. 2021. Regulation (EU) 2021/2115 of the European Parliament and of the Council of 2 December 2021 establishing rules on support for strategic plans to be drawn up by Member States under the common agricultural policy (CAP Strategic Plans) and financed by the European Agricultural Guarantee Fund (EAGF) and by the European Agricultural Fund for Rural Development (EAFRD) and repealing Regulations (EU) No 1305/2013 and (EU) No 1307/2013. Off J Eur Union. 435:1–186.

Ferrarini A, Fracasso A, Spini G, Fornasier F, Taskin E, Fontanella MC, Beone GM, Amaducci S, Puglisi E. 2021. Bioaugmented phytoremediation of metal-contaminated soils and sediments by hemp and giant reed. Front Microbiol. 12:645893. doi:10.3389/fmicb.2021.645893.

Flajšman M, Košmelj K, Grčman H, Ačko DK, Zupan M. 2023. Industrial hemp (*Cannabis sativa* L.) - a valuable alternative crop for growing in agricultural soils contaminated with heavy metals.

Environ Sci Pollut Res Int. 30(54):115414–115429. doi:10.1007/s11356-023-30474-z.
Flemish Government. 2012. Besluit van de Vlaamse Regering van 17 februari 2012 tot vaststelling van het Vlaams reglement betreffende het duurzaam beheer van materiaalkringlopen en afvalstoffen - VLAREMA. Belgian Official Journal.
French Standardization Association. 2006. Standard NF U44-051: organic soil improvers - designations, specifications and marking. Afnor editions.
Gavlak R, Horneck D, Miller RO. 2005. Plant, soil and water reference methods for the western region. 3rd ed. Western Region Extension Publication.
Glivar T, Eržen J, Kreft S, Zagožen M, Čerenak A, Čeh B, Tavčar Benković E. 2020. Cannabinoid content in industrial hemp (*Cannabis sativa* L.) varieties grown in Slovenia. Ind Crops Prod. 145:112082. doi:10.1016/j.indcrop.2019.112082.
Golia EE, Bethanis J, Ntinopoulos N, Kaffe G-G, Komnou AA, Vasilou C. 2023. Investigating the potential of heavy metal accumulation from hemp. The use of industrial hemp (*Cannabis Sativa* L.) for phytoremediation of heavily and moderated polluted soils. Sustain Chem Pharm. 31:100961. doi:10.1016/j.scp.2022.100961.
Hladun KR, Parker DR, Trumble JT. 2015. Cadmium, copper, and lead accumulation and bioconcentration in the vegetative and reproductive organs of *Raphanus sativus*: implications for plant performance and pollination. J Chem Ecol. 41(4):386–395. doi:10.1007/s10886-015-0569-7.
Hossain A, Krupnik TJ, Timsina J, Mahboob MG, Chaki AK, Farooq M, Bhatt R, Fahad S, Hasanuzzaman M. 2020. Agricultural land degradation: processes and problems undermining future food security. In: Fahad S, Hasanuzzaman M, Alam M, Ullah H, Saeed M, Ali Khan I, Adnan M, editors. Environment, climate, plant and vegetation growth. Springer. p. 17–61. doi:10.1007/978-3-030-49732-3_2.
Hou D, O'Connor D, Igalavithana AD, Alessi DS, Luo J, Tsang DCW, Sparks DL, Yamauchi Y, Rinklebe J, Ok YS. 2020. Metal contamination and bioremediation of agricultural soils for food safety and sustainability. Nat Rev Earth Environ. 1(7):366–381. doi:10.1038/s43017-020-0061-y.
Husain R, Weeden H, Bogush D, Deguchi M, Soliman M, Potlakayala S, Katam R, Goldman S, Rudrabhatla S. 2019. Enhanced tolerance of industrial hemp (*Cannabis sativa* L.) plants on abandoned mine land soil leads to overexpression of cannabinoids. PLoS One. 14(8):e0221570. doi:10.1371/journal.pone.0221570.
Inada I, Kiuchi F, Urushihara H. 2023. Comparison of regulations for arsenic and heavy metals in herbal medicines using pharmacopoeias of nine counties/regions. Ther Innov Regul Sci. 57(5):963–974. doi:10.1007/s43441-023-00532-2.
Irshad M, Ruqia B, Hussain Z. 2015. Phytoaccumulation of heavy metals in natural vegetation at the municipal wastewater site in abbottabad, Pakistan. Int J Phytoremediation. 17(12):1269–1273. doi:10.1080/15226514.2014.950409.
Jaros-Tsoj K, Sitko K, Rudnicka M, Sugier P, Jaroszuk-Ściseł J, Rostański A, Rineau F, Papazoglou EG, Alexopoulou E, Vangronsveld J, et al. 2025. Beneficial effects of commercially available preparations of humic substances and mycorrhiza on growth and photosynthesis of sorghum and hemp cultivated on a metal(loid)-polluted field. Plant Soil. 516(2):1553–1574. doi:10.1007/s11104-025-07816-6.
Kashem MA, Singh BR, Kondo T, Imamul Huq SM, Kawai S. 2007. Comparison of extractability of Cd, Cu, Pb and Zn with sequential extraction in contaminated and non-contaminated soils. Int J Environ Sci Technol. 4(2):169–176. doi:10.1007/BF03326270.
Kaur G, Kander R. 2023. The sustainability of industrial hemp: a literature review of its economic, environmental, and social sustainability. Sustainability. 15(8):6457. doi:10.3390/su15086457.
Linger P, Müssig J, Fischer H, Kobert J. 2002. Industrial hemp (*Cannabis sativa* L.) growing on heavy metal contaminated soil: fibre quality and phytoremediation potential. Ind Crops Prod. 16(1):33–42. doi:10.1016/S0926-6690(02)00005-5.
Luyckx M, Blanquet M, Isenborghs A, Guerriero G, Bidar G, Waterlot C, Douay F, Lutts S. 2022. Impact of silicon and heavy metals on hemp (*Cannabis sativa* L.) bast fibres properties: an industrial and agricultural perspective. Int J Environ Res. 16(5):82. doi:10.1007/s41742-022-00446-1.
Marabesi A, Nambeesan S, van Iersel M, Lessl J, Coolong T. 2023. Cadmium exposure is associated with increased transcript abundance of multiple heavy metal associated transporter genes in roots of hemp (*Cannabis sativa* L.). Front Plant Sci. 14:1183249. doi:10.3389/fpls.2023.1183249.
Mark T, Shepherd J, Olson D, Snell W, Proper S, Thornsbury S. 2020. Economic information bulletin 217. Economic viability of industrial hemp in the United States: a review of state pilot programs. Washington DC (USA): US Department of Agriculture - Economic Research Service doi:10.22004/ag.econ.302486.
Meers E, Ruttens A, Hopgood M, Lesage E, Tack F. 2005. Potential of *Brassic rapa*, *Cannabis sativa*, *Helianthus annuus* and *Zea mays* for phytoextraction of heavy metals from calcareous dredged sediment derived soils. Chemosphere. 61(4):561–572. doi:10.1016/j.chemosphere.2005.02.026.
Michels M, Brinkmann A, Mußhoff O. 2025. Economic, ecological and social perspectives of industrial hemp cultivation in Germany: a qualitative analysis. J Environ Manage. 389:126117. doi:10.1016/j.jenvman.2025.126117.
Nicholson FA, Smith SR, Alloway BJ, Carlton-Smith C, Chambers BJ. 2003. An inventory of heavy metals inputs to agricultural soils in England and Wales. Sci Total Environ. 311(1-3):205–219. doi:10.1016/S0048-9697(03)00139-6.
Oeko-Tex. 2025. Oeko-Tex® standard 100: the original safety standard. [accessed date 2026 Jan 28]. https://www.oeko-tex.com/fileadmin/user_upload/Marketing_Materialien/STANDARD_100/Standard/OEKO-TEX_STANDARD100_Standard_EN.pdf
Ofori-Agyemang F, Oustrière N, Waterlot C, Mench M, Burges A. 2026. Field-grown hemp treated with humic/fulvic acids and arbuscular mycorrhizal fungi for phytomanaging a metal-contaminated agricultural soil. Int J Phytoremediation. :1–13. doi:10.1080/15226514.2026.2621118.
Ofori-Agyemang F, Waterlot C, Manu J, Laloge R, Francin R, Papazoglou EG, Alexopoulou E, Sahraoui AL-H, Tisserant B, Mench M, et al. 2024. Plant testing with hemp and miscanthus to assess phytomanagement options including biostimulants and mycorrhizae on a metal-contaminated soil to provide biomass for sustainable biofuel production. Sci Total Environ. 912:169527. doi:10.1016/j.scitotenv.2023.169527.
Pacher P, Kogan NM, Mechoulam R. 2020. Beyond THC and endocannabinoids. Annu Rev Pharmacol Toxicol. 60(1):637–659. doi:10.1146/annurev-pharmtox-010818-021441.
Perlein A, Zdanevitch I, Gaucher R, Robinson B, Papin A, Sahraoui AL-H, Bert V. 2021. Phytomanagement of a metal(loid)-contaminated agricultural site using aromatic and medicinal plants to produce essential oils: analysis of the metal(loid) fate in the value chain. Environ Sci Pollut Res. 28(44):62155–62173. doi:10.1007/s11356-021-15045-4.
Peroni P, Zegada-Lizarazu W, Facciolla E, Parenti A, Mench M, Monti A. 2025. Pairing energy crops and root biostimulants for phytomanaging a former landfill. Ind Crops Prod. 235:121706. doi:10.1016/j.indcrop.2025.121706.
Petrová Š, Benešová D, Soudek P, Van T. 2012. Enhancement of metal(loid)s phytoextraction by *Cannabis sativa* L. J Food Agric Environ. 10(1):631–641. doi:10.1234/4.2012.2735.
Prăvălie R. 2021. Exploring the multiple land degradation pathways across the planet. Earth-Sci Rev. 220:103689. doi:10.1016/j.earscirev.2021.103689.
R Core Team. 2023. R: A language and environment for statistical computing. Ver 4.2.3. [software]. Vienna (Austria): R Foundation for Statistical Computing.
Robinson B, Bañuelos G, Conesa H, Evangelou M, Schulin R. 2009. The phytomanagement of trace elements in soil. Crit Rev Plant Sci. 28(4):240–266. doi:10.1080/07352680903035424.
Saran A, Much D, Vangronsveld J, Merini L. 2024. Phytomanagement of trace element polluted fields with aromatic plants: supporting circular bio-economies. Int J Phytoremediation. 26(2):169–177. doi:10.1080/15226514.2023.2231554.
Shi G, Liu C, Cui M, Ma Y, Cai Q. 2012. Cadmium tolerance and bioaccumulation of 18 hemp accessions. Appl Biochem Biotechnol. 168(1):163–173. doi:10.1007/s12010-011-9382-0.
Smolders E, Oorts K, Van Sprang P, Schoeters I, Janssen CR, McGrath SP, Mclaughlin MJ. 2009. Toxicity of trace metals in soil as affected

by soil type and aging after contamination: using calibrated bioavailability models to set ecological soil standards. Environ Toxicol Chem. 28(8):1633–1642. doi:10.1897/08-592.1.

Sterckeman T, Douay F, Baize D, Fourrier H, Proix N. 2007. Référentiel Pédo-Géochimique du Nord-Pas de Calais. Conseil régional du Nord-Pas-de-Calais.

Sterckeman T, Douay F, Proix N, Fourrier H, Perdrix E. 2002. Assessment of the contamination of cultivated soils by eighteen trace elements around smelters in the north of France. Water, Air, & Soil Pollution. 135(1–4):173–194. doi:10.1023/A:1014758811194.

Struik P, Amaducci S, Bullard M, Stutterheim N, Venturi G, Cromack H. 2000. Agronomy of fibre hemp (*Cannabis sativa* L.) in Europe. Ind Crops Prod. 11(2–3):107–118. doi:10.1016/S0926-6690(99)00048-5.

Sugier P, Jaros-Tsoj K, Jaroszuk-Ściseł J, Papazoglou EG, Alexopoulou E, Sugier D, Sęczyk Ł, Plak A, Kowalski R, Sikorska AD, et al. 2025. Industrial hemp Futura 75 infructescence as a source of essential oils: micromorphology, chemical profile, and yield of essential oils. Ind Crops Prod. 234:121527. doi:10.1016/j.indcrop.2025.121527.

Tack F, Bardos P. 2020. Overview of soil and groundwater remediation. In: Ok YS, Rinklebe J, Hou D, Tsang D, Tack F, editors. Soil and groundwater remediation technologies. 1st ed. CRC Press. p. 1–11. doi:10.1201/9780429322563-1.

Testa G, Corinzia S, Cosentino S, Ciaramella B. 2023. Phytoremediation of cadmium-, lead-, and nickel-polluted soils by industrial hemp. Agronomy. 13(4):995. doi:10.3390/agronomy13040995.

Todde G, Carboni G, Marras S, Caria M, Sirca C. 2022. Industrial hemp (*Cannabis sativa* L.) for phytoremediation: energy and environmental life cycle assessment of using contaminated biomass as an energy resource. Sustain Energy Technol Assess. 52:102081. doi:10.1016/j.seta.2022.102081.

Tóth G, Hermann T, Da Silva MR, Montanarella L. 2016. Heavy metals in agricultural soils of the European Union with implications for food safety. Environ Int. 88:299–309. doi:10.1016/j.envint.2015.12.017.

Trémier A, Buffet J, Daumoin M, Saint Cast P, Picard S, Berger S. 2013. Characterization of digestates: do they fit with soil improvers or fertilisers standards? Paper presented at: the 15th RAMIRAN international conference; Versailles, France.

Uchimiya M, Bannon D, Nakanishi H, McBride M, Williams M, Yoshihara T. 2020. Chemical speciation, plant uptake, and toxicity of heavy metals in agricultural soils. J Agric Food Chem. 68(46):12856–12869. doi:10.1021/acs.jafc.0c00183.

Ure AM. 1990. Trace elements in soil: their determination and speciation. Fresenius J Anal Chem. 337(5):577–581. doi:10.1007/BF00322864.

Van Ranst E, Verloo M, Demeyer A. 1999. Manual for the soil chemistry and fertility laboratory: analytical methods for soils and plants equipment and management of consumables. Ghent (Belgium): NUGI 835.

Vasilou C, Tsiropoulos N, Golia E. 2024. Phytoremediation & valorization of Cu-contaminated soils through *Cannabis sativa* (L.) cultivation: a smart way to produce cannabidiol (CBD) in mediterranean soils. Waste Biomass Valor. 15(3):1711–1724. doi:10.1007/s12649-023-02388-x.

Yin M, Pan G, Tao J, Doblin MS, Zeng W, Pan L, Zhao L, Li Z, Jiang H, Chang L, et al. 2022. Identification of MYB genes reveals their potential functions in cadmium stress response and the regulation of cannabinoid biosynthesis in hemp. Ind Crops Prod. 180:114607. doi:10.1016/j.indcrop.2022.114607.

Zhao X, Guo Y, Papazoglou EG. 2022. Screening flax, kenaf and hemp varieties for phytoremediation of trace element-contaminated soils. Ind Crops Prod. 185:115121. doi:10.1016/j.indcrop.2022.115121.

Zhao X, Wang Y, Wang C, Zhou W, Ouyang D, Gao S, Tan X, Huang R, Guo Y. 2025. Chitooligosaccharide application enhanced the growth and phytoremediation efficiency of industrial hemp in Cd-contaminated soils. Bioresour Technol. 418:131998. doi:10.1016/j.biortech.2024.131998.